Seoyoon Chae

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Profile

Ph.D. Student specializing in AI hardware acceleration, with hands-on experience in Verilog-based ASIC/FPGA design, full digital tape-out flow, and GPU CUDA kernel optimization. Experienced in algorithm-hardware co-optimization, including neural network quantization and KV cache compression for efficient LLM reasoning. Passionate about co-designing ML models and custom accelerator architectures for energy-efficient edge AI.

Education

Rutgers, The State University of New Jersey, Sep. 2026 – Present
Ph.D. in Electrical and Computer Engineering

- **Concentration:** Computer Engineering
- **Advisor:** Prof. Bokyung Kim

SungKyunKwan University (SKKU), B.S. in Electronic and Electrical Engineering Feb. 2022 – Aug. 2026

- **GPA:** 3.89 / 4.5 (Equivalent to 3.65 / 4.0)
- **Coursework:** Digital Systems Design (A+), Digital Integrated Circuit (A+), AI Integrated Circuit Design (A), Electronic & Electrical Programming Lab (A+), Introduction to Machine Learning (A), Parallel Computer Architecture and Programming (A+)
- **Teaching Experience:** Thermodynamics [ISS3317] (Teaching Assistant, Summer 2025)

Publications and Patents

Conference

C2 **Seoyoon Chae**, and Taewook Kang. “LLM on FPGA: Squeezing Language Models by Quantization and Multi-Query Attention and Its Efficient Hardware Architecture,” in *2025 IEEE 22nd International SoC Design Conference (ISOCC)*. (acceptance rate : ∼40%) **[paper] [github]**

C1 Daehee Lee, Sewon Kim, Jungmin Yoo, Yewon Jeong, **Seoyoon Chae**, Chaegyu Lee, Hyunseung Choo, and Jongpil Jeong. “Rotor Fault Diagnosis Method Based on 1D-CNN LSTM,” in *2023 O2O Integrated Conference of the Institute of Internet, Broadcasting and Communication*, pp. 4–6. (in Korean) **[link]**

Patent

P1–P2 Jongpil Jeong, Daehee Lee, Jungmin Yoo, **Seoyoon Chae**, Sewon Kim, Chaegyu Lee, and Yewon Jeong. “Method for Rotor Failure Detection Based on Artificial Intelligence,” Korean Patents 10-2024-0008103 and 10-2024-0008072, filed Jan. 2024 (pending).

Research Experience

BK (BIC, Beyond Intelligent Computing) Lab, Rutgers - *Graduate Research Assistant* Sep.2026 – Present
Advisor: Prof. Bokyung Kim

- `[Sep.2026–Present]` Designed and implemented a DRAM-based Process-in-Memory (PIM) simulator for accelerating dynamic-precision Mixture-of-Experts (MoE) LLM inference.

Efficient and Intelligent Computing (EIC) Lab, SKKU - *Undergraduate Researcher* Jul. 2025 – Aug. 2026
Advisor: Prof. Yulhwa Kim

- `[Dec.2025–Aug.2026]` Developed adaptive KV-cache quantization and tile-level precision control for LLM inference, co-designing quantization policies with memory bandwidth and hardware dataflow constraints to reduce KV memory traffic while preserving model quality.

- `[Jul.2025–Dec.2025]` Built optimized CUDA kernels for LLM inference leveraging block tiling, vectorized memory access, shared-memory reuse, warp-level tiling, and Tensor Core WMMA instructions. Reached near–PyTorch training/inference performance.

Kang Research Group, SKKU - *Undergraduate Researcher* Jul. 2024 – Jul. 2025
Advisor: Prof. Taewook Kang

- `[Jul.2024-Dec.2024]` Fine-tuned CNN keyword-spotting (KWS) models for hardware-friendly deployment using framewise incremental computation scheme; designed and verified Verilog RTL blocks for efficient CNN operations. Participated in the Multi-Project Wafer (MPW) program for undergraduate and graduate students organized by IDEC and contributed to the tape-out of KWS ASIC. **[link]**
- `[Jan.2025-Jun.2025]` Quantized BC-ResNet for keyword spotting(KWS) and speaker-verification(SV) embeddings; implemented Post-Training-Quantization Schemes (CLE, bias correction, AWQ, SmoothQuant). Participated in the Multi-Project Wafer (MPW) program organized by IDEC and contributed to KWS and SV ASIC tape-out. **[link]**
- `[Jun.2025-Jul.2025]` Implemented Multi-Query Attention and W4A4 for FPGA LLM deployment; designed RTL in SystemVerilog for Transformer blocks; Conducted synthesis and bitstream generation for FPGA implementation in Xilinx Vivado. Resulted in publication [C2] **[link]**

Project Experience

Senior Capstone Design, SKKU – Project Leader **[github] [link]** Mar. 2025 – Jun. 2025

- Led a team of four undergraduates to implement and train a CNN-based subpixel rendering algorithm, optimizing network parameters using Incremental Network Quantization scheme.
- Designed its hardware accelerator in Verilog: developed a pipelined structure and validated its functionality.
- Executed the full digital ASIC tape-out flow, including RTL synthesis, timing-driven place-and-route, power and timing sign-off, and final GDSII generation. Awarded Engineering Innovation Prize.

Smart Factory Capstone Design, SKKU – Undergraduate Researcher **[link]** Aug. 2023 – Dec. 2023

- Developed a 1D-CNN LSTM–based fault diagnosis system for rotating machinery using vibration and acoustic signals, achieving up to 97.4% classification accuracy across multiple fault conditions.
- Utilized time-series feature extraction and sequence learning techniques to detect mechanical anomalies such as mechanical looseness, improving predictive maintenance capability. Resulted in [C1] and [P1-2]

Skills

Programming & ML: CUDA (kernel optimization), C/C++, Python (PyTorch)

Hardware & EDA: (System)Verilog, Synopsys IC Compiler II, Design Compiler, Xilinx Vivado, Cadence Virtuoso

Simulation & Verification: gem5, dramsim3, Ramulator 2.0, NVIDIA Nsight Compute, MATLAB

Workshops & Training

- **IDEC Workshop – Synopsys IC Compiler II for Block-Level Auto Place & Route** *Sep. 17–19, 2025*
Hands-on training in ASIC backend design flow including floorplanning, placement, clock tree synthesis (CTS), routing, timing closure, and post-layout verification using Synopsys IC Compiler II.

Awards and Scholarships

School of Engineering Fellow - Rutgers University Sep. 2026 - Jun. 2027

Senior Capstone Design Contest - Engineering Innovation Prize from SKKU Jun. 2025

Extra-Curricular Challenge Scholarship (Top 100 Volunteers in SKKU) Mar. 2025

Academic Excellence Scholarship (Top 12% GPA recipients in SKKU) Fall 2022

Academic Service

- Ad Hoc Reviewer, IEEE Transactions on Circuits and Systems II: Express Briefs (TCAS-II), 2026